

Predictions of Small-Body Physical Traits Based on Orbital Characteristics

Micro-Project 3

[GitHub Repository](https://github.com/Ethereal-Mining-Company/736D616C6C2D626F64792D7633)

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Problem Statement

Small-body diameter prediction methods:

- (1) A combination of very high data collection cost, very high predictive power, and low observational reliability features like albedo (Group 3) with low data collection cost, medium observational reliability features, like absolute magnitude (H) (Group 2)
- (2) A combination of low data collection cost, high observational reliability features (Group 1) with low data collection cost, medium observational reliability features, like absolute magnitude (H) (Group 2)(Ge, 2026; Murcia, 2023; Robinson, 2024).

The project looks to determine if the diameter of a small-body can accurately be predicted with only Group 1 features to maintain the low-cost, high-reliability of features that are inexpensive and not variable or prone to error while observing.

Hypothesis Formulation

The project utilizes a two-model comparison framework that will be assessed from a statistical and operational perspective. The regression models target diameter (Y) using two feature subsets: an expanded set of low-cost, high-reliability observations with low-cost, medium-reliability observations ($\mathbf{X}_{\text{full}}$) and a restricted set of only 'low-cost', 'high reliability' observations ($\mathbf{X}_{\text{restricted}}$).

The restricted model ($\mathbf{X}_{\text{restricted}}$) is preferred due to the lower number and higher reliability of features. Current peer-reviewed research dictates that absolute magnitude (H), a Group 2 feature, is mathematically required to calculate diameter (Ge, 2026).

The hypothesis determines if small-body diameter (Y) can be accurately predicted within a statistical and operational threshold () by the restrictive feature subset compared to the full feature subset, quantified as:

$$\Delta R^2_{\text{test}} = R^2_{\text{Model}(\mathbf{x}_{\text{full}})} - R^2_{\text{Model}(\mathbf{x}_{\text{restricted}})}$$

Hypothesis Formulation

Statistical Hypothesis

The statistical hypothesis tests if the restrictive feature subset of Group 1 features maintains a statistically acceptable degree of accuracy in predicting target diameter (Y) relative to the expanded feature subset of Group 1 and Group 2 features.

- **Statistical Null Hypothesis ($H_{0,stat}$): $\Delta R^2_{test} \leq 0$**
The inclusion of Group 2 features yield no improvement or a decrease in the out-of-sample ΔR^2_{test} , indicating that the expanded features **do not** increase the accuracy of predicting the target diameter (Y).
- **Statistical Alternative Hypothesis ($H_{1,stat}$): $\Delta R^2_{test} > 0$**
The inclusion of Group 2 features yields an increase in the out-of-sample ΔR^2_{test} , indicating that the expanded features **do** increase the accuracy of predicting the target diameter (Y).

Hypothesis Formulation

Operational Hypothesis

The operational hypothesis tests if the restrictive feature subset of Group 1 features maintains an operationally and practically acceptable degree of accuracy () in predicting target diameter (Y) relative to the expanded feature subset of Group 1 and Group 2 features.

- **Operational Null Hypothesis ($H_{0,ops}$):** $\Delta R^2_{test} > \delta$
The inclusion of Group 2 features yield an operationally significant increase in accuracy to predict target diameter (Y), indicating that the restricted feature subset **is not** sufficient enough for operational use.
- **Operational Alternative Hypothesis ($H_{1,ops}$):** $\Delta R^2_{test} \leq \delta$
The inclusion of Group 2 features does not yield an operationally significant increase in accuracy to predict target diameter (Y), indicating that the restricted feature subset **is** sufficient enough for operational use

Acquire

The [JPL NASA Small-Body Dataset](#) was chosen for the following reasons:

- Contains Group 1 and Group 2 characteristics such as absolute magnitude (H), semi-major axis (a), eccentricity (e), and inclination (i) which directly relate to the problem statement and hypothesis requirements necessary for the expanded features subset and restricted feature subset.
- NASA is an authoritative, reliable, and primary repository of small-body data that is also open-source and available to the public allowing reproduction of testing for peer-review.

The [JPL NASA Small-Body Dataset](#) was queried to meet the target variable diameter (Y) to be used in out-of-sample testing. The query exclusively included asteroids with a known diameter greater than zero kilometers (0 km).

Prepare

Data has been prepared for regression analysis in the following ways:

1. **Data Structure Integrity:** Validate all data types are correct.
2. **Data Domain Validation:** Remove data is implausible such as negative diameter.
3. **Duplicate Data Removal:** Remove duplicate rows of data and observations of the same small-body.
4. **Target Variable Isolation (diameter):** Diameter (Y) is the target variable and is required for training.
5. **Exclusion of High-Cost Features (albedo):** Group 3 characteristic are explicitly dropped from the dataset.
6. **Feature Subset Selection:** The model requires the target diameter and features that fall into Group 1.
7. **Feature Completeness:** Entries with missing values in the selected feature subsets were removed.
8. **Target Variable Distribution Scaling:** Scale diameter (Y) using a base-10 logarithmic formula.
9. **Outlier Detection:** Represent real small-bodies and may not be removed.
10. **Feature Independence:** Remove certain highly correlated features.

Prepare

1. Data Structure Integrity

All features are the correct data type. Discrete values like `skpid` (the small-body unique identifier) and `condition_code` are integers (`int`) and all numerical measurements and values like `a` , `e` , and `i` are floats (`float64`).

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 139582 entries, 0 to 139581
Data columns (total 18 columns):
#   Column          Non-Null Count  Dtype
---  -
0   spkid            139582 non-null int64
1   diameter         139582 non-null float64
2   albedo           138459 non-null float64
3   H                139203 non-null float64
4   full_name        139582 non-null str
5   a                139582 non-null float64
6   e                139582 non-null float64
7   i                139582 non-null float64
8   om               139582 non-null float64
9   w                139582 non-null float64
10  q                139582 non-null float64
11  ad               139582 non-null float64
12  per_y            139582 non-null float64
13  data_arc         139577 non-null float64
14  condition_code    139582 non-null int64
15  n_obs_used        139582 non-null int64
16  n_del_obs_used    243 non-null   float64
17  n_dop_obs_used    243 non-null   float64
dtypes: float64(14), int64(3), str(1)
memory usage: 21.7 MB
```

Prepare

2. Data Domain Validation

Removed data that is implausible. Since the data is not validated by an astrophysicist, the project will be limited to checking just the `diameter` (Y). Diameters are all valid because `(df['diameter'] <= 0).sum() = 0`.

```
print(f"Number of invalid diameter records: {(df['diameter'] <= 0).sum()}")
```

Number of invalid diameter records: 0

3. Duplicate Data Removal

Removed rows with identical values in every column and where rows share the same `skpid`. Confirmed removed when `df.duplicated().sum() = 0` and `df.duplicated(subset=['skpid']).sum() = 0`.

```
print("# of rows with identical values in every single column: ", df.duplicated().sum())  
print("# of rows with the same skpid", df.duplicated(subset=['skpid']).sum())
```

of rows with identical values in every single column: 0

of rows with the same skpid 0

4. Target Variable Isolation

All entries without a `diameter` (Y) have been removed. Confirmed because `.isnull().sum() = 0` for `diameter` (Y).

Prepare

5. Exclusion of High-Cost Features

Drop `albedo` because it is a high-cost observation that will not be included in the model. Confirmed removed because feature does not show up in `.info()` or `.isnull().sum()`.

```
df_clean = df.dropna(subset=y).drop(columns=['albedo'])
df_clean.isnull().sum()
```

```
spkid          0
diameter       0
H              379
full_name      0
a              0
e              0
i              0
om             0
w              0
q              0
ad             0
per_y          0
data_arc       5
condition_code  0
n_obs_used     0
n_del_obs_used 139339
n_dop_obs_used 139339
dtype: int64
```

Prepare

6. Feature Subset Selection

Removed all features that are not part of the full feature subset because they are metadata or not orbital characteristics used to determine `diameter` (Y).

```
df_clean = df_clean[[y] + x_full + ['ad', 'q', 'om', 'w']]
df_clean.head()
```

	diameter	a	e	i	H	ad	q	om	w
0	939.400	2.766	0.0797	10.59	3.34	2.99	2.545	80.25	73.29
1	513.000	2.770	0.2307	34.93	4.12	3.41	2.131	172.89	310.97
2	246.596	2.671	0.2557	12.99	5.19	3.35	1.988	169.81	247.90
3	522.770	2.361	0.0902	7.14	3.25	2.57	2.148	103.70	151.47
4	106.699	2.577	0.1876	5.36	6.95	3.06	2.093	141.45	359.38

Prepare

7. Feature Completeness

Checked for any missing or null values using `.isna().sum()`. If missing values, removed the entire entry using `.dropna()`. Since the only feature with missing values is `H` the entries can be deleted without review. This is because `H` is the feature difference in the hypothesis and is fundamental and required to test the model. For the same reasons, it is not appropriate to estimate this value.

```
: df_clean.isna().sum()
```

```
: diameter    0
a             0
e             0
i             0
H            379
ad            0
q             0
om            0
w             0
dtype: int64
```

```
: df_clean.dropna(subset=['H'], inplace=True)
df_clean.isna().sum()
```

```
: diameter    0
a             0
e             0
i             0
H             0
ad            0
q             0
om            0
w             0
dtype: int64
```

Prepare

8. Target Variable Distribution Scaling

Verified `target` `diameter` skew is greater than one (1). Since it was highly skewed, the `diameter` (Y) was transformed into a `log base-10` scale, `diameter_log`, so the regression model does not improperly weight errors between small-body sizes. After the transformation, `diameter_log` was checked for errors in values and the entries were removed as necessary.

```
df_clean['diameter'].describe()
```

```
count    139203.000000
mean         5.469118
std         9.318406
min         0.002500
25%         2.773000
50%         3.957000
75%         5.739000
max        939.400000
Name: diameter, dtype: float64
```

```
print(df_clean['diameter'].skew())
```

```
25.930498594815756
```

```
df_clean['diameter_log'] = np.log10(df_clean['diameter'])
print(f"Total infinity anomalies found: {np.isinf(df_clean['diameter_log']).sum()}")
print(f"Total na values found: {df_clean['diameter_log'].isna().sum()}")
df_clean['diameter_log'].describe()
```

```
Total infinity anomalies found: 0
Total na values found: 0
```

```
count    139203.000000
mean         0.615454
std         0.282694
min        -2.602060
25%         0.442950
50%         0.597366
75%         0.758836
max         2.972851
Name: diameter_log, dtype: float64
```

Prepare

9. Outlier Detection

Calculated the lower and upper bounds from the `interquartile range` (IQR) and reviewed all outliers. No entries were determined to be errors so outliers were kept in the dataset because they represent a real small-body that is part of the population of data. The `diameter` was reviewed to compare to `diameter_log` but will not be used in the model.

```
q1 = df_clean.quantile(0.25)
q3 = df_clean.quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

total_rows = len(df_clean)

for col in df_clean.select_dtypes(include=['number']).columns:
    low_count = (df_clean[col] < lower_bound[col]).sum()
    high_count = (df_clean[col] > upper_bound[col]).sum()

    low_pct = (low_count / total_rows) * 100
    high_pct = (high_count / total_rows) * 100

    print(f"{col} -> Lower Outliers: {low_count} ({low_pct:.2f}%) | Upper Outliers: {high_count} ({high_pct:.2f}%)")

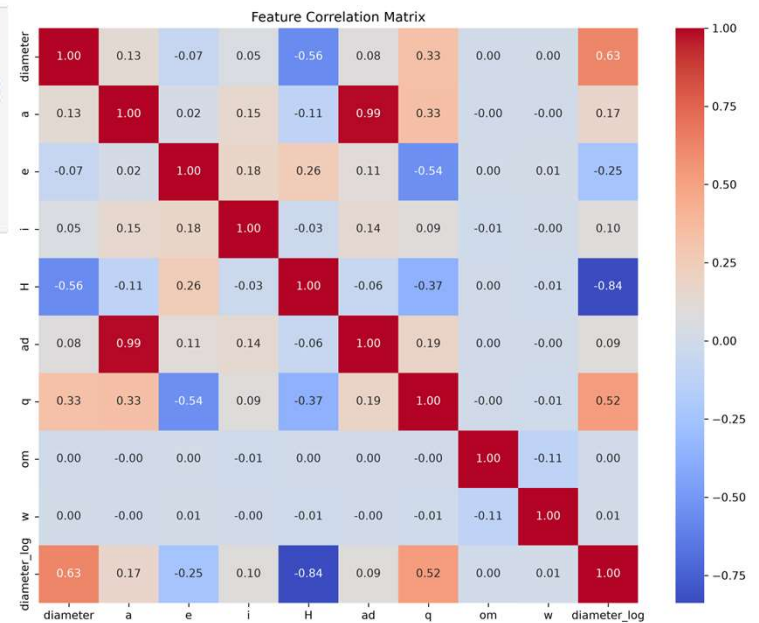
diameter -> Lower Outliers: 0 (0.00%) | Upper Outliers: 9809 (7.05%)
a -> Lower Outliers: 579 (0.42%) | Upper Outliers: 3060 (2.20%)
e -> Lower Outliers: 0 (0.00%) | Upper Outliers: 1886 (1.35%)
i -> Lower Outliers: 0 (0.00%) | Upper Outliers: 4322 (3.10%)
H -> Lower Outliers: 3075 (2.21%) | Upper Outliers: 815 (0.59%)
ad -> Lower Outliers: 449 (0.32%) | Upper Outliers: 3365 (2.42%)
q -> Lower Outliers: 1091 (0.78%) | Upper Outliers: 1983 (1.42%)
om -> Lower Outliers: 0 (0.00%) | Upper Outliers: 0 (0.00%)
w -> Lower Outliers: 0 (0.00%) | Upper Outliers: 0 (0.00%)
diameter_log -> Lower Outliers: 840 (0.60%) | Upper Outliers: 3510 (2.52%)
```

Prepare

10. Feature Independence

Created a heatmap to review highly correlated features. `a` and `ad` are highly correlated. Since `ad` is derived from `a` as $ad = a \times (1 + e)$ it will be removed so it does not get improperly weighted in the regression analysis. Features `om` and `w` have no feature-to-feature correlation or feature-to-target correlation. These will be kept since tree-based regression can factor these out if they are unnecessary.

```
# create correlation matrix to see if any variables inherently correlate to diameter
plt.figure(figsize=(10,8))
sns.heatmap(df_clean.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Feature Correlation Matrix")
plt.tight_layout()
plt.savefig("charts/feature_correlation_matrix.png", dpi=300)
plt.show()
```



Exploratory Data Analysis (EDA)

Data has been analyzed in the following ways:

1. **Individual Feature Boxplots:** Visualize distribution of individual features.
2. **Target Normalization Check:** Visualize log transformation of target.
3. **Primary Signal Feature:** Visualize absolute magnitude (H) versus target.
4. **Orbital Characteristic Features:** Visualize features.
5. **Excluded Features:** Determine features outside of analysis scope.

Exploratory Data Analysis (EDA)

1. Individual Feature Boxplots

Visually reviewed distribution of individual features. Several of the physical features (a , e , and i) are heavily skewed. This indicates linear regression models are not ideal and supports utilizing tree-based models. e (eccentricity) represents the orbit of a small-body with Earth having an $e \approx 0.017$. This means the small-body data set has a relative stable orbit similar to Earth while some outliers have less stable orbits. i represents inclination relative to Earth's (Earth's $i = 0^\circ$). This means the small-body dataset distribution of i has a similar inclination as Earth which makes sense given they are under the same orbital parameters as Earth. a represents the semi-major axis which is the average distance from the Sun. The distribution of a means that most of the small-bodies are in the main asteroid belt between Mars and Jupiter with some outliers as far out as Neptune or farther.

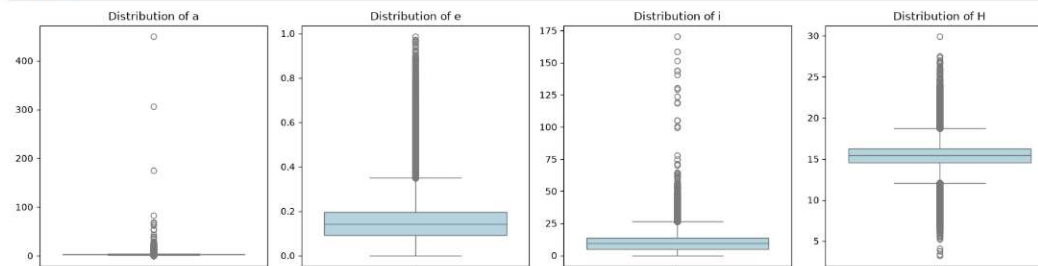
All features have acceptable distributions in context to small-bodies.

```
fig, axes = plt.subplots(2, 4, figsize=(16, 8))
axes = axes.flatten()

for idx, col in enumerate(x_full):
    sns.boxplot(data=df_clean, y=col, ax=axes[idx], color='lightblue')
    axes[idx].set_title(f'Distribution of {col}')
    axes[idx].set_ylabel('')

for i in range(len(x_full), len(axes)):
    fig.delaxes(axes[i])

plt.tight_layout()
plt.savefig("charts/feature_distro.png", dpi=300)
plt.show()
```



Exploratory Data Analysis (EDA)

2. Target Normalization Check

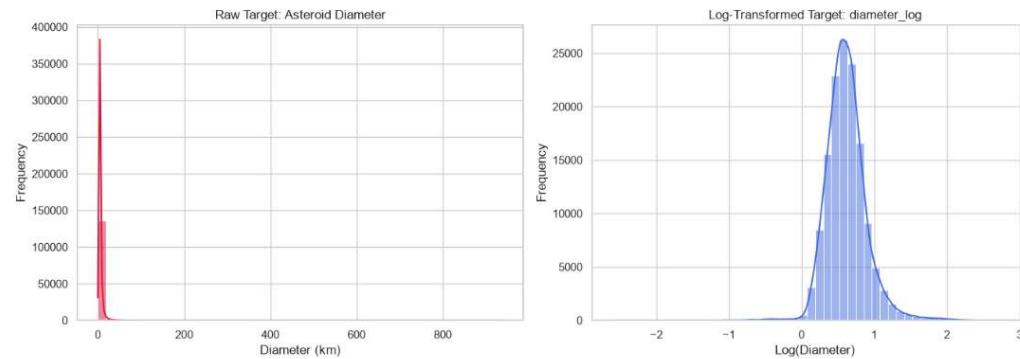
Compared the `diameter` to `diameter_log` to visualize skew. The raw target `diameter` is power-law skewed to the right. The logarithmic transformation decompresses the skew and better prepares the data for the tree-based model analysis.

```
sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.histplot(df_clean['diameter'], bins=50, kde=True, ax=axes[0], color='crimson')
axes[0].set_title('Raw Target: Asteroid Diameter')
axes[0].set_xlabel('Diameter (km)')
axes[0].set_ylabel('Frequency')

sns.histplot(df_clean['diameter_log'], bins=50, kde=True, ax=axes[1], color='royalblue')
axes[1].set_title('Log-Transformed Target: diameter_log')
axes[1].set_xlabel('Log(Diameter)')
axes[1].set_ylabel('Frequency')

plt.tight_layout()
plt.savefig("charts/target_log_transform.png", dpi=300)
plt.show()
```



Exploratory Data Analysis (EDA)

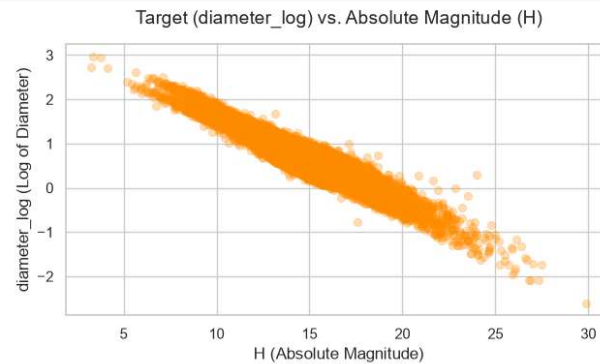
3. Primary Signal Feature

H is the signal feature that differentiates the full and restricted feature subsets and is highly correlated to diameter. The scatterplot visually confirms this correlation and shows an inverse relationship between H and diameter_log. Absolute magnitude (H) represents intrinsic brightness of a small-body where a smaller H value means higher brightness. This trend shows that a small-body with a higher brightness (lower H value) tends to be larger and a small-body with a lower brightness (higher H value) tends to be smaller. This visualization supports the peer-reviewed research that H is highly correlated to diameter and important to predicting a small-body diameter.

```
sns.set_theme(style="whitegrid")
plt.figure(figsize=(6.5, 4))

sns.scatterplot(data=df_clean, x='H', y='diameter_log', alpha=0.3, color='darkorange', edgecolor=None)
plt.title('Target (diameter_log) vs. Absolute Magnitude (H)', fontsize=14, pad=15)
plt.xlabel('H (Absolute Magnitude)', fontsize=12)
plt.ylabel('diameter_log (Log of Diameter)', fontsize=12)

plt.tight_layout()
plt.savefig("charts/target_v_abs_mag", dpi=300)
plt.show()
```



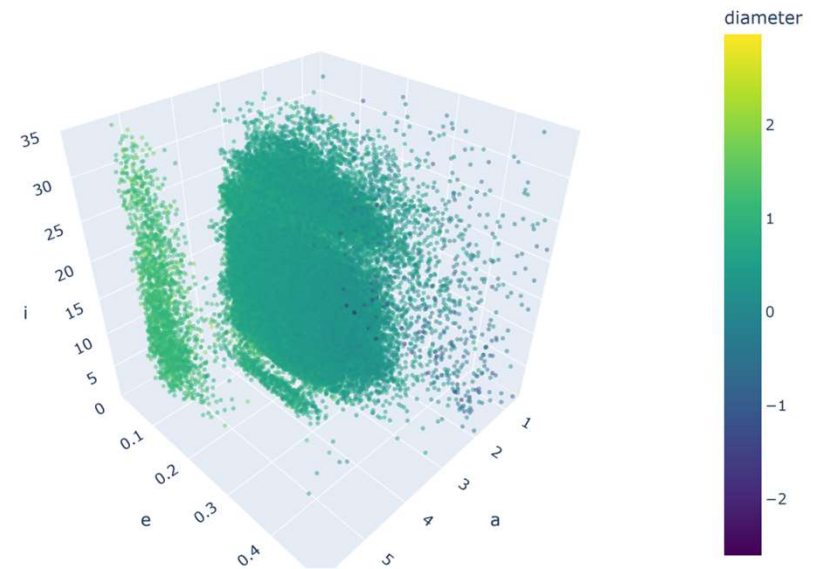
Exploratory Data Analysis (EDA)

4. Orbital Characteristic Features

Fundamental Orbital Elements

Visualizing `a`, `e`, and `i` (independent orbital characteristics) vs. `diameter` to see if there is a correlation between these three features and `diameter`. The visualization reveals distinct clustering demonstrating that certain orbital groupings share diameter profiles. This provides reasonable evidence to use these features as candidates for out-of-sample testing in the model.

Orbital Size, Shape, and Tilt vs Diameter



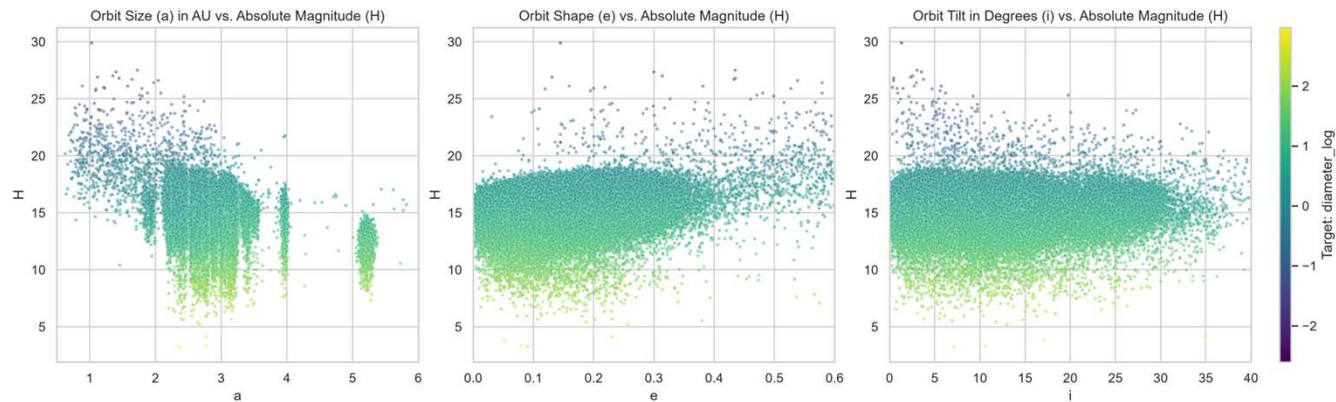
Exploratory Data Analysis (EDA)

4. Orbital Characteristic Features

Absolute Magnitude (H) and Restricted Subset Fundamental Characteristics

Absolute Magnitude (H) versus Individual Restricted Subset Fundamental Characteristics

Visualize a , e , and i individually vs. H in terms of diameter_log . Visualization shows a clear indication that H is a primary indicator for predicting diameter but no distinct correlation exists between H and any of the individual features.



Exploratory Data Analysis (EDA)

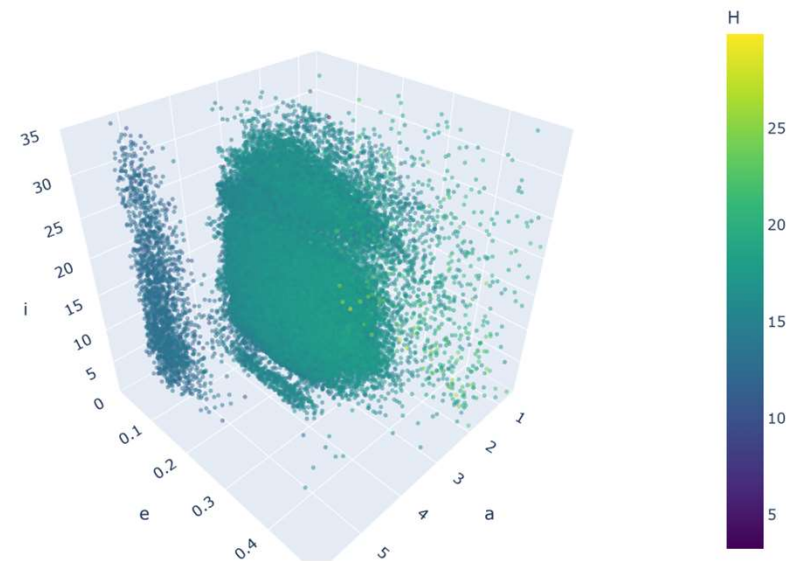
4. Orbital Characteristic Features

Absolute Magnitude (H) and Restricted Subset Fundamental Characteristics

Absolute Magnitude (H) in Relation to Restricted Subset Fundamental Characteristics

Visualizing a , e , and i (independent orbital characteristics) vs. absolute magnitude (H) to see if there is a correlation between these three features and H . The visualization reveals distinct clustering demonstrating that certain orbital groupings share magnitude profiles. This makes sense since H and diameter are closely related so the independent orbital characteristics should exhibit a nearly identical structural trend when mapped by H as they do when mapped by diameter.

Orbital Size, Shape, and Tilt vs Diameter



Exploratory Data Analysis (EDA)

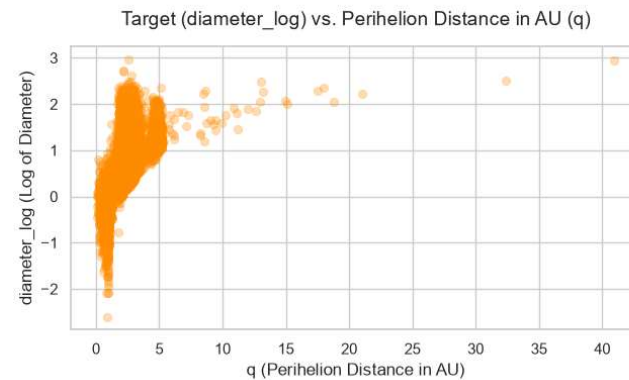
Derived Orbital Characteristics

q is a derived characteristic from a and e as $q = a(1-e)$. Visualizing q vs diameter_log shows a trend that the farther from the Sun a small-body gets, the larger it gets. It has a strong linear slope specifically at 1 AU up to 6 AU. Logically this does not make sense since the size of a small-body is not dictated by the distance from the Sun. There are small and large small-bodies near and far from the Sun. Since this is a **bias** in interpretation and not a flaw in the data, q can be kept in the dataset similar to om and w to be included or excluded in the final model determined at a later step.

```
sns.set_theme(style="whitegrid")
plt.figure(figsize=(6.5, 4))

sns.scatterplot(data=df_clean, x='q', y='diameter_log', alpha=0.3, color='darkorange', edgecolor=None)
plt.title('Target (diameter_log) vs. Perihelion Distance in AU (q)', fontsize=14, pad=15)
plt.xlabel('q (Perihelion Distance in AU)', fontsize=12)
plt.ylabel('diameter_log (Log of Diameter)', fontsize=12)
plt.tight_layout()

plt.savefig("charts/diameter_log_v_peri_dist.png", dpi=300)
plt.show()
```

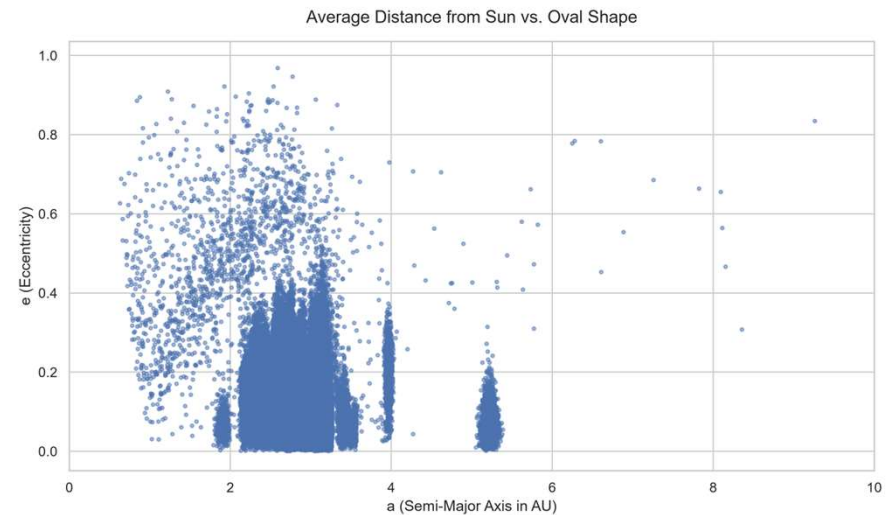


Exploratory Data Analysis (EDA)

4. Orbital Characteristic Features

Semi-Major Axis v. Eccentricity

Visualizing a and e shows us the relationship between the average distance from the Sun and the orbital shape. The visualization does not show a relationship between a and e because there is not a linear relationship. The visualization does show that small-bodies tend to hold specific orbital distances with a major segment between 2 and 3.5 AU and another smaller segment around 5.2 AU. This is consistent with the main asteroid belt between Mars and Jupiter and a smaller one at Jupiter's orbit.

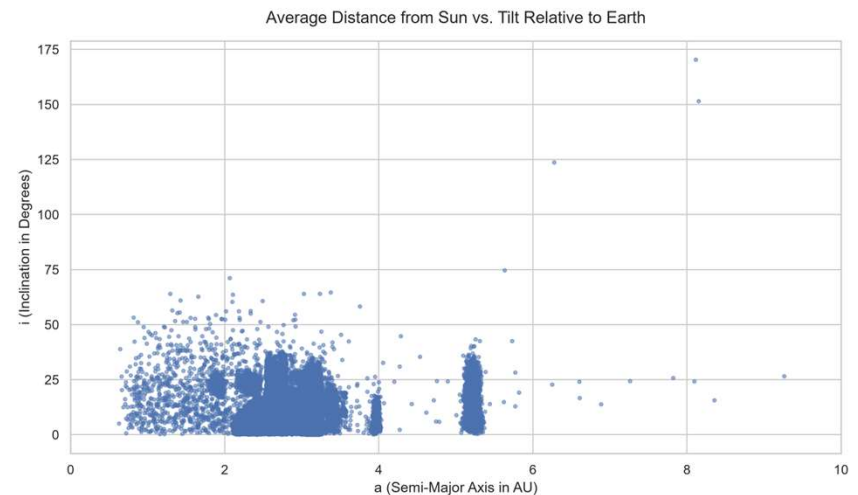


Exploratory Data Analysis (EDA)

4. Orbital Characteristic Features

Semi-Major Axis v. Inclination

Visualizing a and i shows us the relationship between the average distance from the Sun and the inclination relative to Earth. The visualization does not show a relationship between a and i because there is not a linear relationship. The visualization does show that small-bodies throughout the inner solar system tend to have the same inclination as Earth.

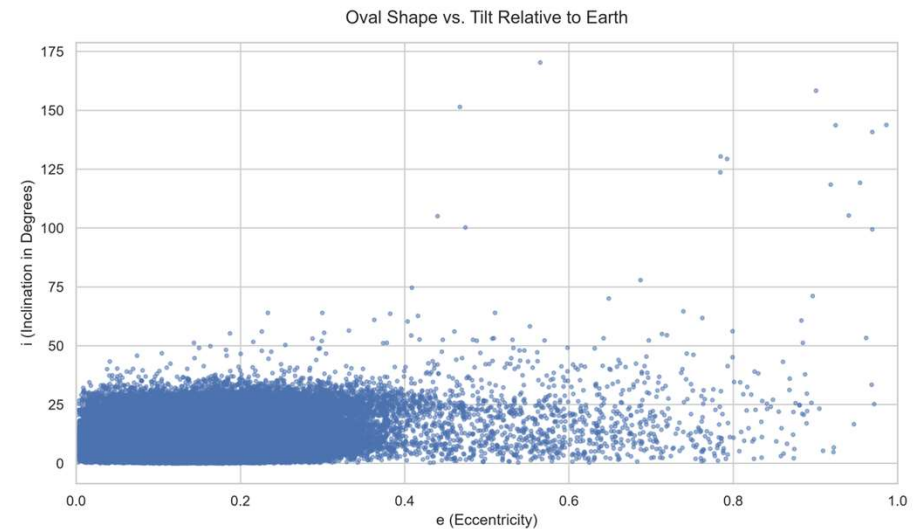


Exploratory Data Analysis (EDA)

4. Orbital Characteristic Features

Eccentricity v. Inclination

Visualizing e and i shows us the relationship between the orbital shape and the elliptical plane. The visualization does not show a correlation between e and i because there is not a linear relationship.



Exploratory Data Analysis (EDA)

5. Excluded Features

om and w were excluded from visualization as the feature correlation matrix determined there was not a strong correlation between these features and other features or the target. While these features may or may not introduce noise to the tree-based regression models, peer-reviewed analysis has determined that they do not provide strong predictive indicators for diameter. Due to the computing power and time to analyze additional features in the dataset, these features will only be analyzed as an ad hoc if time permits prior to the project deadline.

Exploratory Data Analysis (EDA)

1. EDA Insights

Preliminary analysis of the dataset shows that the key indicators of a small-body size (target diameter (Y)) are absolute magnitude (H), semi-major axis (a), eccentricity (e), and inclination (i). There is a strong non-linear relationship between diameter and H indicating that H is a strong predictor of diameter. While there were no linear correlations between individual a, e, i features and diameter or H, comparison of a, e, and i mapped to diameter and a, e, and i mapped to H provided similar results. This provides strong evidence that a, e, and i could meet the operational threshold to predict diameter without the need for H. The EDA alone is not enough to conclude if the null hypothesis or the alternative hypothesis are correct, but there is a pattern and relationship between a, e, i, and diameter.

Exploratory Data Analysis (EDA)

2. Feature Determination

Analysis of the features indicates that a, e, and i are strong candidates for the regression model because they have a non-linear, structural relationship to the target. The feature q is reserved for future determination for inclusion or exclusion in the regression model since it is derived from a and e which could introduce multicollinearity or noise. Features om and w can be excluded. While om and w are not derived from any other features, the lack of correlation to the target may introduce noise to the regression model that could lead to overfitting.

Exploratory Data Analysis (EDA)

3. Model Selection

An analysis of the dataset and feature determination support using a random forest regression model to test the hypothesis. Small-bodies form dense structural clustering in the feature space and a highly skewed target distribution. Small-bodies also can be observed at great distances which creates mathematical outliers. Physical characteristics and derived characteristics may have no correlation to the size of a small-body or introduce multicollinearity. A random forest regression model uses piecewise step-function approximations to handle extreme outliers and noise of derived or irrelevant features. This is better than linear regression or other regression models that try to fit data into a straight line or curve. This will result in stronger models to increase accuracy to predict small-body sizes and to test the hypothesis.

Analysis

```
--- [Run 1/4 Training Random Forest (Default Parameters) ---

=====
INITIALIZING UNIFIED RANDOM FOREST PIPELINE
=====
Dataset Path:    data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:   (139203, 7)
Full Features:    ['a', 'e', 'i', 'H']
Restrictive Features: ['a', 'e', 'i']
Target Column:    diameter_log
Model Parameters: {'n_estimators': 150, 'max_depth': 25, 'min_samples_split': 5, 'random_state': 42, 'n_jobs': -1}
=====

Running RANDOM FOREST Cross-Validation Evaluation...
Fold 1/5 Complete | Full R²: 0.8492 | Restricted R²: 0.3319 | Δ R²: 0.5173
Fold 2/5 Complete | Full R²: 0.8514 | Restricted R²: 0.3360 | Δ R²: 0.5154
Fold 3/5 Complete | Full R²: 0.8513 | Restricted R²: 0.3325 | Δ R²: 0.5188
Fold 4/5 Complete | Full R²: 0.8473 | Restricted R²: 0.3321 | Δ R²: 0.5152
Fold 5/5 Complete | Full R²: 0.8549 | Restricted R²: 0.3327 | Δ R²: 0.5222
Running RANDOM FOREST production pipeline...

=====
CROSS-VALIDATION SUMMARY AVERAGES (RANDOM FOREST)
=====
Full Model Mean R²:      0.8508
Restricted Model Mean R²: 0.3330
Mean Δ R² (Drop Impact): 0.5178 (± 0.0026)
Full Model Mean MAE:      1.0442 km
Restricted Model Mean MAE: 2.4091 km
=====

Training final production RANDOM FOREST pipelines on full dataset...
Success! RANDOM FOREST production bundles saved safely (Old binaries overwritten).

===== HYPOTHESIS TESTING REPORT (RANDOM FOREST) =====
Testing against Operational Tolerance (δ) = 0.05
Calculated Delta R² (Drop Impact):      = 0.5178
-----
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
=====

History logged automatically under [RF] with UUID: 47096e74-a06e-4e5d-aaba-851469d94bf8
```

Analysis

```
--- [Run 2/4 Training Random Forest Variant (Higher Sample Splits) ---

=====
INITIALIZING UNIFIED RANDOM FOREST PIPELINE
=====
Dataset Path:      data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:     (139203, 7)
Full Features:     ['a', 'e', 'i', 'H']
Restrictive Features: ['a', 'e', 'i']
Target Column:     diameter_log
Model Parameters:  {'n_estimators': 150, 'max_depth': 25, 'min_samples_split': 10, 'random_state': 42, 'n_jobs': -1}
=====

Running RANDOM FOREST Cross-Validation Evaluation...
Fold 1/5 Complete | Full R²: 0.8510 | Restricted R²: 0.3401 | Δ R²: 0.5109
Fold 2/5 Complete | Full R²: 0.8529 | Restricted R²: 0.3446 | Δ R²: 0.5083
Fold 3/5 Complete | Full R²: 0.8529 | Restricted R²: 0.3414 | Δ R²: 0.5115
Fold 4/5 Complete | Full R²: 0.8490 | Restricted R²: 0.3413 | Δ R²: 0.5077
Fold 5/5 Complete | Full R²: 0.8564 | Restricted R²: 0.3416 | Δ R²: 0.5148
Running RANDOM FOREST production pipeline...

=====
CROSS-VALIDATION SUMMARY AVERAGES (RANDOM FOREST)
=====
Full Model Mean R²:      0.8524
Restricted Model Mean R²: 0.3418
Mean Δ R² (Drop Impact): 0.5106 (± 0.0026)
Full Model Mean MAE:     1.0417 km
Restricted Model Mean MAE: 2.3926 km
=====

Training final production RANDOM FOREST pipelines on full dataset...
Success! RANDOM FOREST production bundles saved safely (Old binaries overwritten).

===== HYPOTHESIS TESTING REPORT (RANDOM FOREST) =====
Testing against Operational Tolerance (δ) = 0.05
Calculated Delta R² (Drop Impact):      = 0.5106
-----
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
=====

History logged automatically under [RF] with UUID: ddbbb103-a46d-4f88-a721-f05cef295ddf
```

Analysis

```
--- [Run 3/4 Training XGBoost (Default Parameters) ---

=====
INITIALIZING UNIFIED XGBOOST PIPELINE
=====
Dataset Path:      data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:     (139203, 7)
Full Features:      ['a', 'e', 'i', 'H']
Restrictive Features: ['a', 'e', 'i']
Target Column:     diameter_log
Model Parameters: {'n_estimators': 150, 'max_depth': 8, 'learning_rate': 0.05, 'min_samples_leaf': 2, 'random_state': 42}
=====

Running XGBOOST Cross-Validation Evaluation...
Fold 1/5 Complete | Full R²: 0.8527 | Restricted R²: 0.3582 | Δ R²: 0.4945
Fold 2/5 Complete | Full R²: 0.8547 | Restricted R²: 0.3606 | Δ R²: 0.4941
Fold 3/5 Complete | Full R²: 0.8557 | Restricted R²: 0.3659 | Δ R²: 0.4899
Fold 4/5 Complete | Full R²: 0.8524 | Restricted R²: 0.3652 | Δ R²: 0.4872
Fold 5/5 Complete | Full R²: 0.8584 | Restricted R²: 0.3653 | Δ R²: 0.4931
Running XGBOOST production pipeline...

=====
CROSS-VALIDATION SUMMARY AVERAGES (XGBOOST)
=====
Full Model Mean R²:      0.8548
Restricted Model Mean R²: 0.3630
Mean Δ R² (Drop Impact): 0.4918 (± 0.0028)
Full Model Mean MAE:      1.0360 km
Restricted Model Mean MAE: 2.3461 km
=====

Training final production XGBOOST pipelines on full dataset...
Success! XGBOOST production bundles saved safely (Old binaries overwritten).

===== HYPOTHESIS TESTING REPORT (XGBOOST) =====
Testing against Operational Tolerance (δ) = 0.05
Calculated Delta R² (Drop Impact):      = 0.4918
-----
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
=====

History logged automatically under [XGBOOST] with UUID: e7aa49fa-fb1b-44a3-a087-7ba9e11bcc9f
```

Analysis

```
--- [Run 4/4 Training XGBoost Variant (Lower Learning Rate) ---

=====
INITIALIZING UNIFIED XGBOOST PIPELINE
=====
Dataset Path:      data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:     (139203, 7)
Full Features:      ['a', 'e', 'i', 'H']
Restrictive Features: ['a', 'e', 'i']
Target Column:     diameter_log
Model Parameters:  {'n_estimators': 200, 'max_depth': 8, 'learning_rate': 0.04, 'min_samples_leaf': 2, 'random_state': 42}
=====

Running XGBOOST Cross-Validation Evaluation...
Fold 1/5 Complete | Full R²: 0.8532 | Restricted R²: 0.3586 | Δ R²: 0.4945
Fold 2/5 Complete | Full R²: 0.8553 | Restricted R²: 0.3607 | Δ R²: 0.4946
Fold 3/5 Complete | Full R²: 0.8555 | Restricted R²: 0.3656 | Δ R²: 0.4899
Fold 4/5 Complete | Full R²: 0.8524 | Restricted R²: 0.3652 | Δ R²: 0.4872
Fold 5/5 Complete | Full R²: 0.8582 | Restricted R²: 0.3660 | Δ R²: 0.4923
Running XGBOOST production pipeline...

=====
CROSS-VALIDATION SUMMARY AVERAGES (XGBOOST)
=====
Full Model Mean R²:      0.8549
Restricted Model Mean R²: 0.3632
Mean Δ R² (Drop Impact): 0.4917 (± 0.0028)
Full Model Mean MAE:      1.0341 km
Restricted Model Mean MAE: 2.3459 km
=====

Training final production XGBOOST pipelines on full dataset...
Success! XGBOOST production bundles saved safely (Old binaries overwritten).

===== HYPOTHESIS TESTING REPORT (XGBOOST) =====
Testing against Operational Tolerance (δ) = 0.05
Calculated Delta R² (Drop Impact):      = 0.4917
-----
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
=====

History logged automatically under [XGBOOST] with UUID: c4dcd1ea-81a7-4f6f-8063-1c0e4942422e
```

Analysis

Run Timestamp	Model UUID Token	Engine Type	Delta Threshold	Calculated ΔR^2	Statistical Outcome	Operational Outcome	Full R^2	Restricted R^2	Delta Std CV R^2	Full MAE (km)	Restricted MAE (km)	Model Parameters
10/04/26 20:41	47096e74-a06e-4e5d-aaba-851469d94bf8	RF	0.0500	0.5178	H1_stat	H0_ops	0.8508	0.3330	0.0026	1.0442	2.4091	{"n_estimators": 150, "max_depth": 25, "min_sa...
10/04/26 20:45	ddb1b103-a46d-4f88-a721-f05cef295ddf	RF	0.0500	0.5106	H1_stat	H0_ops	0.8524	0.3418	0.0026	1.0417	2.3926	{"n_estimators": 150, "max_depth": 25, "min_sa...
10/04/26 20:58	e7aa49fa-fb1b-44a3-a087-7ba9e11bcc9f	XGBOOST	0.0500	0.4918	H1_stat	H0_ops	0.8548	0.3630	0.0028	1.0360	2.3461	{"n_estimators": 150, "max_depth": 8, "learnin...
10/04/26 21:15	c4dcd1ea-81a7-4f6f-8063-1c0e4942422e	XGBOOST	0.0500	0.4917	H1_stat	H0_ops	0.8549	0.3632	0.0028	1.0341	2.3459	{"n_estimators": 200, "max_depth": 8, "learnin...

Analysis

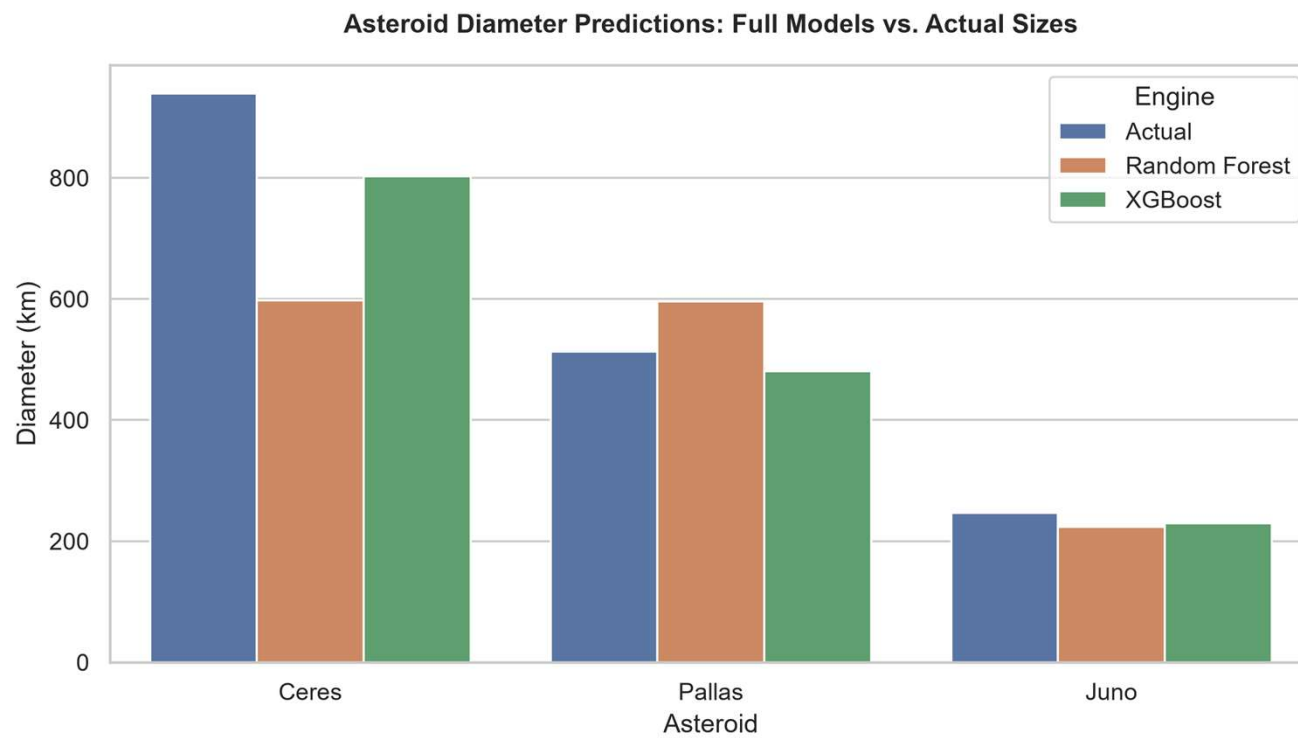
OVERALL PREDICTION SUMMARY TABLE:

Asteroid	Actual Diameter (km)	Engine	Full Pred (km)	Full Error %	Restricted Pred (km)	Restricted Error %
Ceres	939.4000	Random Forest	597.4500	36.4000	22.3100	97.6000
Ceres	939.4000	XGBoost	802.7800	14.5000	4.2200	99.6000
Pallas	513.0000	Random Forest	595.9100	16.2000	23.8400	95.4000
Pallas	513.0000	XGBoost	480.3000	6.4000	23.4200	95.4000
Juno	246.5960	Random Forest	223.6200	9.3000	11.8400	95.2000
Juno	246.5960	XGBoost	229.8000	6.8000	5.2700	97.9000

Mean Absolute Percentage Error (MAPE) by Engine for Full Model:

Engine	MAPE
Random Forest	20.63%
XGBoost	9.23%

Analysis



Report

Hypothesis Testing

Acceptance of **Statistical Alternative Hypothesis ($H_{1,stat}$)**: $\Delta R^2_{test} > 0$

The inclusion of Group 2 features yields an increase in the out-of-sample ΔR^2_{test} , indicating that the expanded features **do** increase the accuracy of predicting the target diameter (Y).

- All four models resulted in a ΔR^2_{test} of no less than 0.4917. This exceeded the statistical threshold of zero (0). This means the expanded features did increase the accuracy of predicting the target diameter (Y).

Acceptance of **Operational Null Hypothesis ($H_{0,ops}$)**: $\Delta R^2_{test} > \delta$

The inclusion of Group 2 features yield an operationally significant increase in accuracy to predict target diameter (Y), indicating that the restricted feature subset **is not** sufficient enough for operational use.

- All four models resulted in a ΔR^2_{test} of no less than 0.4917. This exceeded the operational threshold of 0.05. This means the restricted feature subset is insufficient for production deployment.

Report

Model Performance & Selection

Acceptance of Gradient Boosting (gbr) tree-based model over Random Forest (rf) tree-based model. GBR resulted in a MAPE of 9.23% versus RF with a MAPE of 20.63%. This means that GBR missed predicting the true size of the small-body by 9.23% where RF did 20.63%.

GBR was a better model for this use case because it used sequential error correction versus independent averaging like in RF. The ability for GBR to learn from prior tree mistakes, especially in the heavily skewed dataset, made it better at building a model to more accurately predict the target.

Report

Key Insights & Limitations

The training and prediction testing shows that more in-depth and sophisticated training is needed to predict target diameter with both the full and restrictive subset. While the full feature subset models had R^2_{test} values around 0.85, the RF prediction of Ceres is a clear indication that there are still issues with the model. This is likely due to the skewed nature of small-bodies where some are very large and the bulk are very small. This skew makes it hard for even tree-based models to accurately predict small-body diameter with the given feature set.

This is also evident in the MAE of the full feature subset. The values for all models ranged from 1.0341-1.0442 while the predictions were significantly off from these averages. This means that the skew of small-body distribution artificially suppresses the global MAE downward.

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